Curriculum Otherwise - Sight Reading Factory: A Critical Analysis

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The Artifact:

Sight Reading Factory (SRF) is an application that can be used in web browsers and iOS devices. It is primarily designed as a tool for musicians and students to practice "sight reading" – the act of reading or performing a piece of music for the very first time through clapping, singing, or playing an instrument. Sight reading exercises can be generated on SRF by selecting from predefined levels of difficulty or by creating custom exercises with control over parameters such as rhythm, range, key signature, time signature, and tempo. There are also different playback sounds from piano, voice, and guitar to instruments from the strings, woodwind, brass, and percussion families. Other features of SRF that can be accessed from the main hub are a "Practice Log" where students can keep track of and stay accountable for their learning, and gamified "Quests" that show attendance streaks, achievements, and time-limited challenges.

The new media aspect of SRF comes from how this technology combines standardized Western music notation with generative algorithmic composition. This affordance shifts the relationship between the student and the act of sight reading, in which the student is enabled to practice their music fluency by working through a continuous stream of generated sight reading material. This platform can also function as a classroom platform for assessment where students can, for example, be sorted into groups and be asked to record themselves performing assigned exercises with set parameters for teacher feedback of pitch and rhythmic accuracy.

SRF is primarily marketed toward music educators and musicians, and like any tool, it carries an implied curriculum, a set of competencies that it assumes any user to have: the ability to read and subdivide standard rhythms; the ability to hold a steady pulse across different and changing tempos; and the ability to put these skills together to read an unfamiliar series of rhythms at performance speed. These aren't stated learning outcomes but they are what SRF requires to use it. These assumptions usually fit the profile of someone who has gone through an elementary music program, but nothing is said about how the tool should handle a student who hasn't yet developed these preliminary music literacy skills.

Research Literature:

Reifinger (2020) explains that notation reading develops sequentially, moving from an embodied, aural grasp of sound (the "enactive" stage) to the abstract visual symbol (the "symbolic" stage). While his focus is on pitch, the same logic applies to rhythm: audiation must precede reliable symbolic reading. This gives us a developmental standard against which to read SRF's implied curriculum, in which SRF's settings assume the symbolic stage is already in place, with no mechanism for confirming the enactive stage underneath it.

Stenberg and Cross (2019) found that vertical white gaps marking sub-phrasal units act as a cognitive chunking device, significantly reducing sight-reading errors. This gives us an empirical standard for evaluating the readability of SRF's generated notation and the cognitive load it places on students, an angle the course readings have not equipped us to assess so far. This also helps us read against the expectation that SRF has for its users – that they should be able to successfully sight read the entirety of a generated passage – which may be overwhelming for still-developing readers.

Kalantzis and Cope (2010) argue that learning and self-motivation occurs best when students feel a sense of belonging to their learning environment and to the content that is taught, which can only be done if they are included. This perspective makes us appreciate the flexibility that SRF offers in terms of its customizable settings when generating exercises, while still allowing us to be critical of which identities are left out of taking full advantage of these affordances and who might therefore feel alienated; examples include students with no experience reading notation at all to students with visual or motor impairments.

The experiment conducted by Bossavit and Parsons (2018) showed that gamification could provide sustained engagement and motivation in a controlled, non-threatening learning environment for students diagnosed with autism spectrum disorder (ASD). There were observations into how engagement differed with the content and with each other when students cooperated and competed against each other. While the authors noted the limitations of their findings to generalize to the larger population of individuals with ASD, this study is a good starting point for us to consider how SRF might be used effectively in a classroom setting when there are fluid factors to consider such as social dynamics and students with neurodivergent approaches to learning.

Graham (2024) discusses the advantages of developing sight reading skills in a group setting and having peers working with and helping each other – either formally through a pairing or mentorship system, or informally through observation and practice with each other. Though the attention is on melodic sight reading in choral ensembles, the attention brought to the combination of sight reading strategies with sight reading exercises may be invaluable to a wide range of students, including but not limited to: syllable systems to vocalize rhythms (e.g. Takadimi) and reading notation in context to the surrounding music as opposed to scanning notes one at a time. If implemented in conjunction with SRF, these strategies could combat the assumption of users having sight reading proficiency while also giving them a stepping stone to help them eventually reach said proficiency.

What We Are Seeing:

Putting Sight Reading Factory and the BC Music curriculum side by side makes apparent that both carry an implicit idea of who the student is and where their abilities should currently be at. The settings on SRF are calibrated for a user who already has a relationship to staff notation and is refining a skill rather than building one from nothing. D'Ignazio and Klein's (2020) concept of the "privilege hazard" names this mismatch: a tool's design can quietly assume a student who does not match the one actually in the room, treating the dominant or normative experience as the default.

This raises the question of who gets left out. There is the student who joined a band or strings program for the first time in middle school without having the opportunity to experience general music in elementary school. There is the student whose rhythmic sense comes from an aural tradition, a garage band, or a family musical practice rather than from staff notation. Neither the BC curriculum nor SRF excludes these students outright, but neither plans for them either. SRF lets users jump right into practice, but this comes at the cost of having no scaffolding for the beginner student on how to read rhythms in the first place. Ọnụọha's (2016) concept of "missing datasets," developed to describe the data that is not collected about marginalized groups, is useful here by analogy: the students left out represent the blank spots in a system that treats them as an exception rather than a possibility worth designing for.

Alvarez's (2019) critique of "technology-enhanced learning" sharpens this further. It marks the difference between a curriculum that is sequenced and developmental where students are explicitly asked to use creative processes to "explore relationships between identity, place, culture, society, and belonging" (British Columbia Ministry of Education and Child Care, n.d.), and a tool like SRF that generates material on demand. Drawing on the work of Paulo Blikstein, Alvarez warns that such tools often foster "learned passivity" by treating students as consumers of software rather than active builders of knowledge (Blikstein, 2008, as cited in Alvarez, 2019). SRF can render a rhythm harder or easier, but it cannot determine what a Grade 8 student needs to consolidate in simple meters before a Grade 9 student advances to the more complex compound or mixed meters.

Finally, Gillespie (2010) gives us a way to think about the new media dimension specifically: a platform's design hides its politics behind a neutral appearance. SRF is not simply digitizing an existing practice but changing what sight reading practice is, rendering it infinite, generated, and recorded for grading rather than scarce, chosen, and private. What we have not yet worked out is how much that shift matters at this particular age when private practice is becoming visible to a teacher or an algorithm may change who feels comfortable inside the practice at all.

Our Emerging Critical Stance:

Our stance toward this artifact is currently one of interrogation with some pull toward extension. We are not arguing that SRF should be discarded or that it has no place in a music classroom, but rather questioning the assumptions built into it, particularly around who it naturally serves as the “ideal user” and who it leaves behind. Our own recognition of the mismatch between the tool's intended user and the actual students in a BC classroom is itself an instance of working through “privilege hazard” (D'Ignazio & Klein, 2020). As musicians ourselves, we are able to engage with the tool and its affordances, but this renders its exclusions invisible.

MusicFirst will be donating licenses so we can carry out this project, and we want to be clear that this support does not change our stance. We are looking at SRF critically and noting what is missing rather than simply showcasing what it does well, and we are equally invested in how the tool could be changed to be more accessible, inclusive, and acknowledging of the ever-changing classroom environment.

We are drawn toward extension because the underlying idea of generating endless rhythm practice material on demand carries real potential for a Grade 8/9 program. That potential depends on pairing it with a more deliberate sense of sequence and a stronger connection to the repertoire students are actually rehearsing. Our position, then, is less about rejecting SRF and more about asking what it would take to make its underlying idea actually fit the students the BC curriculum says it is for. This means SRF needs us to keep examining its built-in assumptions before we add anything new to it.

Gesturing Toward Design:

Our design gestures seek to actively challenge SRF's structural constraints by making deliberate choices about what to keep and what to refuse. This includes:

1. **Building in a developmental sequence.** Drawing on Alvarez (2019), we reject the assumption that a student should simply adapt to whatever difficulty setting SRF provides. We want the tool's settings mapped onto the BC Music curriculum's own grade-by-grade language and into a logical sequence so that simple meters are strengthened before compound and mixed meters are introduced.
2. **Connecting practice to repertoire.** We like the idea of generating fresh rhythm material on demand, since that solves a real problem of running out of practice material. We disagree that sight reading should be disconnected from the repertoire that ensembles are learning. We want a way to seed SRF's generated rhythms from the rhythmic figures already in a class's current pieces, so practice becomes meaningful and personal, and also reinforces the music students are working on instead of running parallel to it.
3. **Widening who the tool sees as a student.** SRF's classroom tools manage the ensemble, not the individual, and offer little for a student with a visual or motor-related impairment beyond what a generic touchscreen can offer. We want to keep the efficiency that group-level assignment gives a teacher when managing a full class, but we refuse the idea that the ensemble is the only unit worth designing for. This means building in alternative ways for a student to show rhythmic understanding, whether it is through the physical actions of clapping back a rhythm or call and response, or by using adaptive peripheral technology to allow more inclusion. That way, notation will not be the only path to being perceived as competent.
4. **Building in corrective feedback, not just scoring.** Right now SRF's feedback marks a performance right or wrong without explaining what to do differently or offering tips and strategies, which shows what the platform registers as a pedagogical choice, not a neutral one as Gillespie (2010) points out. Students

benefit from knowing how they're doing, but we refuse a model that uses a binary of pass/fail scoring. Instead, we propose feedback that is more active. Drawing on Reifinger's (2020) enactive/symbolic distinction, we would like to open up more opportunities for students, like giving them a chance to clap or audiate a missed rhythm before reading it again, or for the software to give suggestions for breaking down the rhythm and counting it.

5. **Designing for access, not just skill level.** We want to keep the goal of making rhythm practice widely available, since that's a real strength of the tool. But there is an assumption that every student already has a device and reliable internet connection at home which is a limitation of the tool that we reject; following Ọnụọha (2016), those students are a missing dataset the tool wasn't built to notice. We'd want generated exercises exportable as printable worksheets, so practice can occur “offline” as well.

Jeremy’s Connection: When I was teaching band, I never had the budget or resources to bring technology like SRF into my classroom. That left me outside the user SRF imagined, the same gap this paper's access argument is built on, just lived from the teacher's side rather than the student's. That history is also what makes this paper's access argument feel concrete to me rather than abstract, in a way it might not be for someone who already had the tools, abilities and drive to become fluent. My own relationship to reading notation has been a fickle one. I struggled with sight-reading on guitar, especially holding a steady flow without stopping to fix mistakes or knowing how something would sound before I played it, which is close to the struggle this paper names between the enactive and the symbolic (Reifinger, 2020). What pulls me toward SRF's potential is its control over the reading experience itself. I used to lose focus with several pages of music in front of me at once, and I'm curious whether stripping away that clutter would make it more accessible to stay with what’s being sight-read as the music moves on. I'm still working out where I land on SRF, but I want to know whether its rhythms and pitches can be generated in the style of real composers and etudes, and whether the algorithm understands music theory well enough to guarantee the quality of what it generates.

Chris's Connection: As a music teacher and someone who has spent twenty years performing in the local jazz, orchestral and wind ensemble community, my position gives me a very practical, firsthand look at the friction between technology and live performance. On the bandstand, music is entirely relational and collaborative; you learn by audiating, listening and adjusting to the musicians around you. When I look at SRF through that lens, I am hyper-aware of how the tool can isolate an individual student behind a screen, trading a shared cultural experience for automated, point-based grading. Because this software aligns with my own musical background, I experienced what D'Ignazio and Klein (2020) call a "privilege hazard", meaning the tool naturally worked for me, which initially made its structural exclusions invisible.

Yet, my reality as a classroom teacher means I cannot simply dismiss the software. Managing large secondary ensembles involves massive administrative hurdles. Practically speaking, SRF solves a grueling workflow problem for me; it allows me to integrate sight reading on the fly without spending days compiling, formatting and printing physical booklets like I have had to do in the past. Furthermore, I have found real success using it collectively by projecting the generated notation for a full ensemble warm-up. I often turn it into a gamified routine with the whole class, pushing them to play a generated line perfectly before we move on, throwing in jokes about being "so close" when they almost nail a tricky section. By changing how the software is physically delivered, I am altering what Gillespie (2010) identifies as the platform's hidden politics, transforming an isolated testing mechanism into a shared instructional routine that keeps the practice communal and lighthearted. My position puts me in a daily balancing act. I am constantly trying to leverage its massive logistical convenience and group warm-up benefits without letting its algorithmic tracking turn music education into a sterile checklist that leaves non-traditional learners behind.

Samuel's Connection: I was never able to engage with SRF while studying music as the paywall was a significant barrier, even though I had heard stories about its effectiveness. I had a natural affinity for reading notation and for audiation, accompanied by severe performance anxiety which made it difficult to translate that knowledge into my stiff fingers in real-time when playing the piano. I knew I had to improve my sight reading skills to become a better musician, but had no idea where to look without having to spend more money on pieces that I could only truly sight read once. As a result, my learning was impacted by my lack of sense of belonging to the music community, as Kalantzis & Cope describe (2010), and I almost left it all behind. Now that I am in the teaching profession and working with the elementary population, I believe I can revisit SRF as a jumping-off point for the introduction and reinforcement of sight reading notation in a more informal environment, using its gamified features to make learning music fun while lessening feelings of anxiety and of falling behind, something I am all too familiar with. The ability to customize the difficulty of the exercises brings adaptability so that I can meet students where they are, and SRF's generative ability means that educators will never have to worry about running out of material, a severe limitation of relying on physical notation to practice with continually.

References

Alvarez, M. (2019). (Digital) media as critical pedagogy. *Media Theory, 3*(1), 73–102. https://doi.org/10.70064/mt.v3i1.964

Bossavit, B., & Parsons, S. (2018). Outcomes for design and learning when teenagers with autism codesign a serious game: A pilot study. *Journal of Computer Assisted Learning*, *34*, 293–305. https://doi.org/10.1111/jcal.12242

British Columbia Ministry of Education and Child Care. (n.d.). *Arts education: Music, kindergarten to grade 9.* Building Student Success – B.C. Curriculum. https://curriculum.gov.bc.ca/curriculum/arts-education

D'Ignazio, C., & Klein, L. F. (2020). *Data feminism.* MIT Press.

Gillespie, T. (2010). The politics of 'platforms'. *New Media & Society, 12*(3), 347–364. https://doi.org/10.1177/1461444809342738

Graham, M. F. (2024). The effects of peer-assisted learning on rhythmic and melodic sight-reading in a middle school chorus. *International Journal of Music Education*. https://doi.org/10.1177/02557614241288957

Kalantzis, M. & Cope, B. (2010). The teacher as designer: Pedagogy in the new media age. *E-learning and Digital media*, *7*(3), 200-222. https://doi.org/10.2304/elea.2010.7.3.200

Ọnụọha, M. (2016). *The library of missing datasets* [Installation]. https://mimi-onuoha-9s0o.squarespace.com/the-library-of-missing-datasets

Reifinger, J. L., Jr. (2020). Teaching pitch notation–reading skills. *General Music Today, 33*(3), 21–28. https://doi.org/10.1177/1048371319891419

Stenberg, A., & Cross, I. (2019). White spaces, music notation and the facilitation of sight-reading. *Scientific Reports, 9*, Article 5299. https://doi.org/10.1038/s41598-019-41445-1